



**MINISTRY OF TECHNOLOGY AND SCIENCE**  
*Industrial Training Centre  
Management Board*



**DIPLOMA IN COMPUTER SYSTEMS ENGINEERING**

NAME : XXXXXXXXXXXXXXXX  
ID NUMBER : XXXXXXXXXX  
PROGRAM : DCSE  
COURSE : COMPUTING PROJECT  
LECTURER : MR MUBANGA  
TOPIC : XXXXXXXXXXXXXXXX  
TASK : CHAPTER 4 FINAL PROJECT REPORT

## **System Guide/Manual**

### **Introduction**

The Online Student Loans and Bursary Web Application is an application to automate the processes involved in applying for scholarships and bursaries among student applicants who may wish to apply for sponsorship. It targets students from all parts of the country allowing them to complete their applications online or through the internet.

### **Who should use this manual?**

This manual is specific to the tasks that will be performed by student applicants who may wish to apply for scholarships.

### **Process overview**

The web application admin is there to insert, update, and delete the data entered by users in the database connected to the application.

All the information received from users through the forms is managed by the web application admin.

### **Browsing the web app through home page**

To browse the web application, click on the drawer to navigate certain options that will explain about the purpose of the app. Navigate the home page by clicking on buttons such as Login, Sign, about, and Contacts. You can further explore other categories and search for specific options.

To navigate other buttons for specific purposes, click on the drawer and open the drawer to navigate available scholarships, available bursaries, loan repayment, and how to play.

Login in the Online Student Loans and Bursary Web Application, the user of the web app has to click on the button indicating **Login** to enter email and password.

On the login screen, the user login through credentials given. After the login, navigate the web and fill in the given form to apply.

Upon successful Login the following web page will open and the user will click on apply to proceed with the application.

## **Creating an Account**

To create an account, click on the “**Sign**” button at the top right of the page. Fill in the required information including your name, email address, and password.

## **Receiving applications**

- i. Applicants will submit their applications the online application form.
- ii. Applications will be stored in the database and can be only viewed by logged-in users.
- iii. An email notification is set up to alert users when new applications are received.

## **Managing applications**

- i. View and manage applications by clicking on the “Applications” button.
- ii. Filter applications by level of qualifications or other criteria.
- iii. Review applicant information, including cover letter and other documents.
- iv. Update status of each application.

## **Communication with Applicants**

- i. Send automated email responses to applicants upon submission.
- ii. Use the Online Student Loans and Bursary Web Application to send personalized emails to applicants.

## **Reporting and Analytics**

- i. Generates reports on application metrics such as number of applications received.
- ii. Use analytics to identify challenges faced by applicants.

## **Security and Data Protection**

- i. Our Online Student Loans and Bursary Web Application uses industry-standard security measures to protect applicant data.
- ii. Ensure that only authorized users have access to the Online Student Loans and Bursary Web Application and applicant information.

By following this guide, you will be able to effectively use our Online Student Loans and Bursary Web Application to receive and manage applications from applicants. If you need any further assistance, please do not hesitate to contact our team.

## **Installation Manual**

### **System Requirements**

- ✓ Database (MySQL)
- ✓ Web server (Apache)
- ✓ Programming languages (Html, PHP, Python, JavaScript)

### **Installation Steps**

- i. **Download the web application**  
Download the web application from the provided link or repository.
- ii. **Extract the files**  
Extract the downloaded files to a directory on your web server.
- iii. **Configure the database**  
Create a new database and update the web application's configuration file with the database credentials.
- iv. **Install dependencies**  
Install any required dependencies using the package manager (composer or pip).
- v. **Run the installation script**  
Run the provided installation script to set up the web application.
- vi. **Configure the web server**  
Configure the web server to the point to the web application's directory.
- vii. **Test the installation**  
Test the web application to ensure it is working correctly.

### **Post-installation configuration**

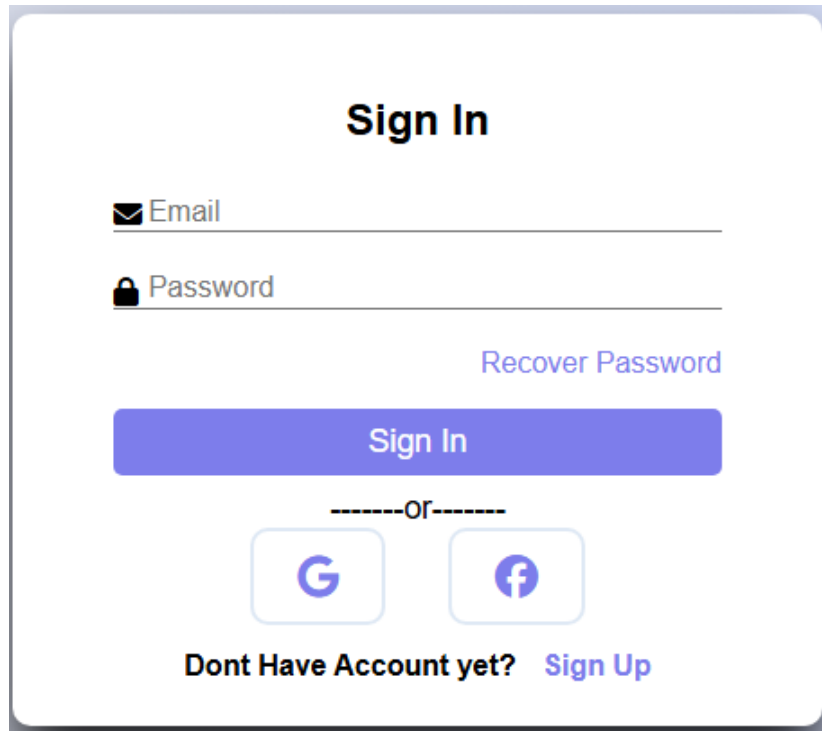
- i. **Set up administration account**  
Create an administrator account to manage the web application.
- ii. **Configure email settings**  
Configure email settings to send notifications to applicants.

By following these installation steps and post-installation configuration, you will have a fully functional web application receiving applications from applicants. If you encounter any issues, contact our support.

## Testing Plan, Output Plan

### Application submission (Case 1 steps)

- i. Login or signup with your credentials to be granted permission to proceed with application.



**Sign In**

✉ Email

🔒 Password

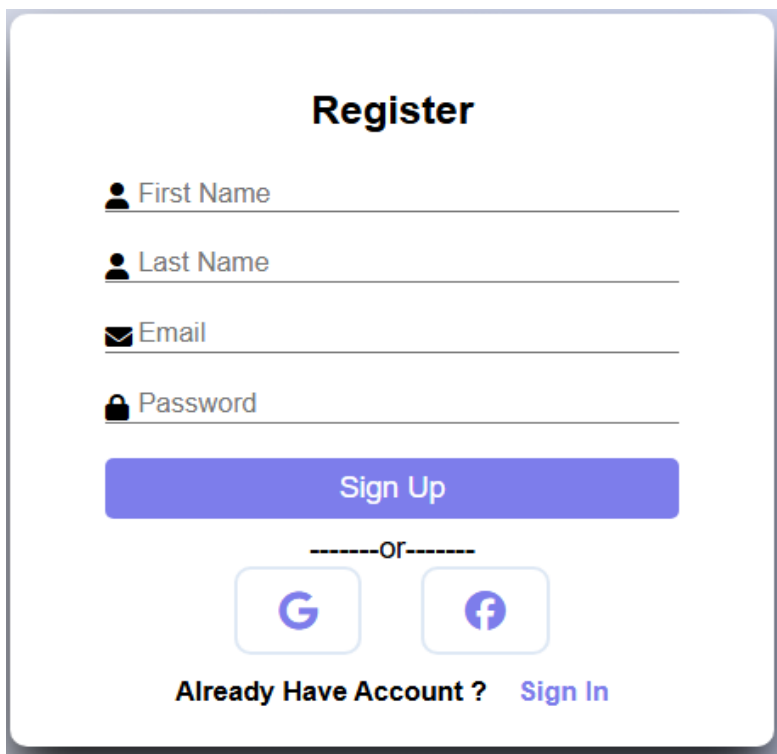
[Recover Password](#)

**Sign In**

-----Or-----

**Dont Have Account yet?** [Sign Up](#)



**Register**

 First Name

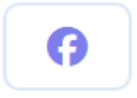
 Last Name

✉ Email

🔒 Password

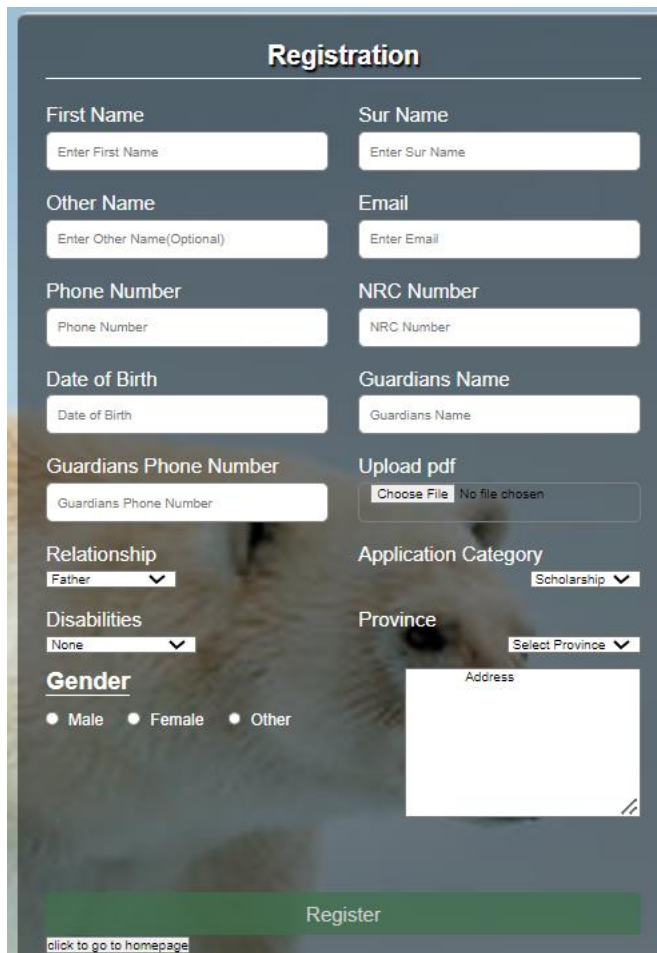
**Sign Up**

-----Or-----

**Already Have Account ?** [Sign In](#)

- ii. Fill out application form with correct details required in the form.
- iii. Attach all the required documents.
- iv. Submit application (Expected result: “Application is submitted successfully and confirmation message is displayed”).



The image shows a web-based registration form titled "Registration". The form is divided into two columns and includes the following fields and controls:

- First Name:** Text input field with placeholder "Enter First Name".
- Sur Name:** Text input field with placeholder "Enter Sur Name".
- Other Name:** Text input field with placeholder "Enter Other Name(Optional)".
- Email:** Text input field with placeholder "Enter Email".
- Phone Number:** Text input field with placeholder "Phone Number".
- NRC Number:** Text input field with placeholder "NRC Number".
- Date of Birth:** Text input field with placeholder "Date of Birth".
- Guardians Name:** Text input field with placeholder "Guardians Name".
- Guardians Phone Number:** Text input field with placeholder "Guardians Phone Number".
- Upload pdf:** File upload control with a "Choose File" button and "No file chosen" text.
- Relationship:** Dropdown menu with "Father" selected.
- Application Category:** Dropdown menu with "Scholarship" selected.
- Disabilities:** Dropdown menu with "None" selected.
- Province:** Dropdown menu with "Select Province" as the placeholder.
- Gender:** Radio button group with options "Male", "Female", and "Other".
- Address:** Text area for entering the address.

At the bottom of the form, there is a green "Register" button and a link that says "click to go to homepage".

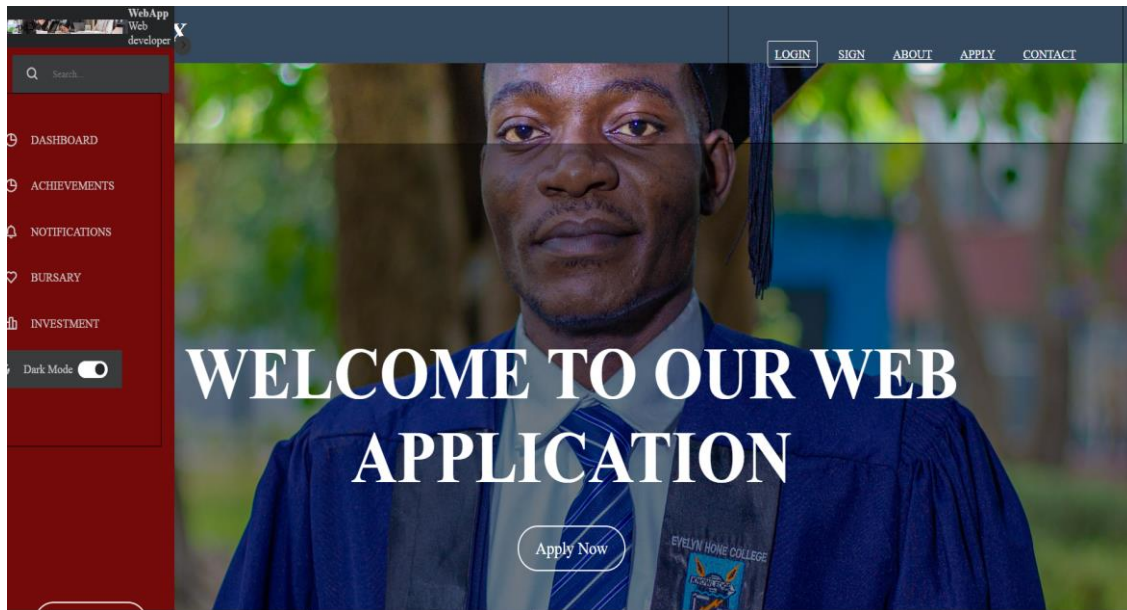
### Application submission (Case 2 steps)

- i. Fill out the form with invalid data (e.g. missing required fields).
- ii. Attempt to submit application (Expected result: “error message is displayed and application is not submitted”).

### Application viewing (View submitted application steps)

- i. Log in as administrator
- ii. View submitted application (Expected result: “Application details are displayed correctly”)
- iii. View application PDF (Expected result: “Application PDF is generated correctly”).

## Home page



The home page shows the first layout of our project and a number of buttons that given to navigate the system. The drawer represents the shortcuts to help users navigate without challenges and without taking time.

### Application status (Update application status steps)

- i. Log in as administrator
- ii. Update application status (Expected result: “Application status is updated successfully”).

### Communication (Send Application Confirmation Email steps)

- i. Submit application
- ii. Verify confirmation email is sent (Expected result: “Confirmation is sent successfully”).

### Security (Validate Password Strength steps)

- i. Attempt to create account with weak password
- ii. Verify error message is displayed (Expected result: “Error message is displayed and account is not created”).

## Main function codes

### //Sign Up code

```
<form method="post" action="special.php">
  <div class="input-group">
    <i class="fas fa-user"></i>
    <input type="text" name="fName" id="fName" placeholder="First Name" required>
    <label for="fname">First Name</label>
  </div>
  <div class="input-group">
    <i class="fas fa-user"></i>
    <input type="text" name="lName" id="lName" placeholder="Last Name" required>
    <label for="lname">Last Name</label>
  </div>
  <div class="input-group">
    <i class="fas fa-envelope"></i>
    <input type="email" name="email" id="email" placeholder="Email" required>
    <label for="email">Email</label>
  </div>
  <div class="input-group">
    <i class="fas fa-lock"></i>
    <input type="password" name="password" id="password" placeholder="password" required>
    <label for="password">Password</label>
  </div>
  <input type="submit" class="btn" value="Sign Up" name="signUp">
..
```

### //Log in code

```
<form method="post" action="special.php">
  <div class="input-group">
    <i class="fas fa-envelope"></i>
    <input type="email" name="email" id="email" placeholder="Email" required>
    <label for="email">Email</label>
  </div>
  <div class="input-group">
    <i class="fas fa-lock"></i>
    <input type="password" name="password" id="password" placeholder="password" required>
    <label for="password">Password</label>
  </div>
  <p class="recover">
    <a href="#">Recover Password</a>
  </p>
  <input type="submit" class="btn" value="Sign In" name="signIn">
```



## **Summary**

In summary, we have discussed chapter four of our final project proposal and we have looked at parts of the chapter sequentially. We have talked about the system and manual guide of our web application where we have stressed on how our user interface is designed and how the user can navigate the web app, a user will quickly see the login and signup buttons upon opening the web. We have highlighted the steps to follow when one wants to create an account or signup, how applications will be received from applicants, how to manage these applications, how to communicate with applicants through emails, how to analyse and help applicants, and how to our web application is secured in terms of data protection. We have further discussed the installation manual where we have focused on explaining the system requirements such as the database and programming languages to use, we highlighted the installation steps for our web application and lastly we have provided some of the main function codes of our web application.